

Nikhil Gupta

B.Tech – Electronics and VLSI Design



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Education

Dhirubhai Ambani University <i>CPI: 8.08</i>	2024 - Present Gandhinagar, Gujarat
Neehar Meera Senior Secondary School (CBSE) <i>Percentage: 83.8%</i>	2022 - 2024 Aligarh, Uttar Pradesh
Delhi Public School (CBSE) <i>Percentage: 87.8%</i>	2018 - 2022 Aligarh, Uttar Pradesh

Projects

Radix Converter — Web + FPGA <i>JavaScript, HTML/CSS, Verilog, Vercel</i>	GitHub Try it Out
• Built a dual-platform radix (base) converter — a static web app deployed on Vercel and an independent Verilog implementation on a Digilent Nexys A7 FPGA — sharing the same conversion algorithm. • Engineered exact BigInt fraction arithmetic (numerator/denominator) instead of floating point, so non-terminating fractions are truncated at 20 digits and visibly marked rather than silently rounded. • Implemented automatic input-base detection with an explicit user-confirmation step, preventing silent misreads when a number typed in one base is interpreted as another. • Verified the FPGA design in Icarus Verilog across unit, protocol, and end-to-end levels, catching a real UART busy-signal timing bug that silently dropped characters during back-to-back transmission.	
AI Resume Chatbot <i>Python, FastAPI, Pydantic, Groq LLM API, pypdf</i>	GitHub Try it Out
• Built a two-stage LLM pipeline that extracts unstructured PDF resume text into a validated, nested Pydantic schema using schema-guided JSON-mode prompting, eliminating manual data entry. • Engineered a grounded Q&A layer that injects the parsed resume as structured JSON into the system prompt with strict anti-hallucination rules, so the model answers only from candidate data. • Designed a schema-first architecture where a single Pydantic model drives both LLM output validation and the prompt contract, removing hand-written parsing and regex extraction. • Exposed the pipeline via a FastAPI REST service with typed request/response models and OpenAPI docs; reduced per-query latency and API cost by caching the parsed resume and cutting LLM calls from two to one per message.	

Technical Skills

Languages: C, C++, Python, Verilog
Developer Tools: Git, VS Code, Linux Shell, Figma, Claude
Soft Skills: Team Collaboration, Leadership, Problem Solving, Communication, Time Management
Relevant Coursework: Data Structures and Algorithms, Real-Time Operating Systems, Signals and Systems
Areas of Interest: Full Stack Development, System Design, Competitive Programming, AI/ML, Cloud Computing

Position Of Responsibility

Deputy Convener, Headrush (Quizzing Club) Dhirubhai Ambani University	2025 – Present Gandhinagar, Gujarat
• Organized and managed inter-college quizzing events, coordinating logistics, sponsorships, and participant outreach.	

Achievements & Extracurriculars

Smart India Hackathon (SIH) National Level Hackathon	Ongoing
* Developing a solution for AI-Enabled Antarctic Sea-Ice, Iceberg Trajectory, and Navigation Decision Support System. Cleared the internal college rounds and advanced to the further stage.	